

Linguistic Meaning Requires Stable Concepts

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Abstract

Nicholas Shea’s *Concepts at the Interface* (2024) identifies concepts with temporary working memory (WM) labels: ‘two-ended’ devices connected to stored information at one end and composing with other labels at the other. I argue that this framework cannot yet account for linguistic understanding. First, Shea’s theory requires stable categorical mechanisms to track category identity over time, although he is officially neutral as to their nature. Second, even granting such mechanisms, neither they nor his temporary WM labels can provide the needed compositional vehicles. The empirical evidence I review converges on the need for stable, freely compositional vehicles that I call ‘G-concepts’. On the view developed here, G-concepts—stable symbols individuated by their compositional role—compose into thoughts and bear word meaning, while ‘S-concepts’—stable representations that underwrite categorization—orient the mind toward the world. The Hybrid view preserves Shea’s interface insight while providing the compositional resources needed for linguistic cognition.

concepts, mental representation, compositional semantics, polysemy, propositional memory, language acquisition

1 Introduction

Nicholas Shea’s *Concepts at the Interface* (Oxford University Press, 2024) proposes a radical reconceptualization of concepts in cognitive science. On the classical computational view à la Fodor, a concept is a stable mental representation that both composes into thoughts and gives access to the information used in categorization. Shea separates these roles: composition is handled by occurrent working memory (WM) labels, categorization by information stored in long-term memory.

Unlike Fodor’s *Concepts*, however, Shea’s book engages very little with linguistic cognition. This is where I shall press him. My central claim is that something like (1) is true, but that Shea’s framework lacks the resources to supply the needed concept F .

- (1) For any expression E , there is a particular concept F involved in a competent speaker’s grasp of the meaning of E .

(1) is intended to be neutral among various metasemantic options. I am drawn to an identity reading, on which F simply is the meaning of E (cf. Fodor 1998, p. 2), but nothing below depends on it.

§2 sets out Shea’s picture in his terms. §3 says what (1) demands of F , and introduces the Hybrid view. My argument then proceeds in two steps. Shea’s framework requires stable categorical mechanisms that track category identity over time, despite his official neutrality about what individuates a store (§4). Granting those, linguistic competence furthermore requires stable compositional vehicles—what I call G-concepts—to bear word meanings and compose into thoughts (§5). Such vehicles persist, freely compose, and are ready for reasoning even when speakers know little to nothing about what they apply to.

2 Shea’s picture

Shea’s book aims to unify two research programs. On one side, cognitive scientists have extensively studied special-purpose systems—those serving vision, motor control, numerical cognition, folk physics, and so on. On the other side, researchers have studied general-purpose, compositional thought—our ability to freely combine concepts to reason about anything whatsoever.

On Shea’s diagnosis, each program is incomplete without the other, and they meet at the theoretical posit ‘concept’:

When I think, *what sport shall I play this week?*, I am making use of my concept SPORT, putting it together with other concepts to form the thought. Concepts then connect our thinking to all of the information, of different kinds, that we rely on in thought. The concept SPORT gives me access to semantic memories (e.g. that ultimate frisbee is an easy sport to get into) and to information in many special-purpose systems: what is physically involved in various activities and how I am likely to feel doing them. (p. 3)

The two sides of the relevant connections differ in format. Special-purpose systems use “a representational system or range of representational models suited to some purposes and not others” (p. 121), with “compositional principles that are different to the general-purpose concatenation of conceptual thought” (p. 121). Conceptual thought, on the other hand, is formatted much as Fodor supposes: “[l]anguage-like compositional structure is plausibly exhibited by the conceptual representations which are tokened in episodes of conscious deliberation”, and the compositional principle at work “is general-purpose, in the sense that any concept can be combined in thought with any other concept or concepts of the right kind” (p. 44).

Each format displays signature reasoning properties. An inference is *content-general*, broadly, when its warrant turns on the logical vocabulary—e.g., inferring from *Bo has a gleebe* to *Bo has something*, whatever gleebs are—and *content-specific* when it turns on the content of the non-logical terms—e.g. as when one infers from something’s being a dog that it has fur (pp. 65–66). Shea doesn’t claim a unique correspondence here, but on his picture content-specific inference is the bread and butter of special-purpose systems, while content-general inference is so for conceptual thought.

The vehicles of thought, in Shea’s framework, are temporary WM labels. Within an episode, a label is a vehicle type, individuated by the non-semantic properties that fix how it is processed (p. 137); across episodes, labels “do not correspond to a single mental word (vehicle type)” (p. 125), and what makes one a token of a concept is what it is connected to. This separation of the label from what it labels is motivated by a conflation that Shea diagnoses in the literature, between the representation tokened in thought and the store of information accessible from it:

On the one hand, [concepts] are a representation type, a representation of X, that is tokened in occurrent thinking. On the other hand, they are a store of information about Xs that is used in categorisation, inference, and other cognitive processes. Many theorists have assumed that the same representation plays both roles... [But] the representations in working memory displaying general-purpose compositionality that we reason with in deliberation are not themselves bodies of information that are used in cognition. (p. 125)

Fodor is offered as the paradigm of what is rejected: “[f]or Fodor a concept is a mental word that serves to give access to stored information. The same word, *qua* vehicle type, shows up on each occasion” (p. 141). Shea’s concepts

are instead two-ended, “plug-and-play” devices: a concept interfaces “between the informational models and content-specific computations of special-purpose systems, at one end, and the general-purpose compositionality and content-general reasoning of deliberate thought, at the other” (p. 123). Officially, the concept just is the WM label: “[i]n short, a concept is a working memory label” (p. 152). Shea considers counting the label-plus-retrieved-characterizations as the concept instead, and declines (“I prefer the former”), while insisting that type-identification runs through label and store either way (p. 217). A ‘store’ for dogs is the comprehender’s information about dogs, held in long-term memory “in both general-purpose and special-purpose informational models” (p. 141); thinking about dogs involves connecting a WM label to that store.

Since WM labels are temporary, the identity of a concept across episodes of thinking runs through the store:

Labels deployed on different occasions are instances of the same concept-type, in that thinker, just in case they are connected to the same store of information (p. 138).

Shea does not individuate concepts by the information at a store (dodging holism; p. 139), but by the store itself, which is said to persist through changes in contents much like Theseus’s ship (p. 138). But Shea is careful not to commit on what, if anything, holds a store together: his picture, he tells us, is compatible with there being no index at all (p. 152).

3 The problematic, and the Hybrid view

In my view, states that can serve as the F s in (1)—‘concepts’—face three requirements. They must be stable: it must be possible for the same word to map to the same concept across contexts and occasions of use, lest we lose any baseline against which meaning multiplicity phenomena like indexicality, homonymy, and polysemy are recognized as departures (see Gillon 2004 for diagnostics). They must be compositional, figuring by rule in any of the boundlessly many thoughts that we can express with sentences. And they must be about something.

I accept Shea’s separation of roles between the representations composed in higher-level thinking and the information deployed when categorizing things in experience. I ask what fills each role. As Shea notes, Fodor distinguishes “the atom that enters into computations and the store of information to which it is connected”; the difference is that, for Fodor, the occurrent vehicle type also indexes the store (p. 141). Shea’s stores persist, but they do not compose; his

WM labels compose, but they do not persist. Yet, Shea’s ground for denying persisting vehicle types concerns access, not composition; he elsewhere allows that “[t]he same label-vehicle might, on different occasions, be used to label different stores” (p. 138).

An alternative, Hybrid view makes persistent vehicles available for thought (G-concepts), as on Fodor’s view, but separates them from the (first-order) representations we deploy as categories (S-concepts).

G-concepts (General-purpose concepts): Discrete mental symbols in a general-purpose, amodal format that combine freely with others up to the limits of their syntactic compatibility, meeting the Generality Constraint (cf. Evans 1982; Camp 2004). Stable, stored, and available across episodes of thinking. Directly support content-general inferences, figure in speakers’ grasp of word meaning, and compose into the thoughts expressed by sentences.

S-concepts (Special-purpose concepts): Mental representations that track categories and underwrite classification and categorization behavior. Stable and available across episodes of thinking, like G-concepts, but different in kind: they need not be and often are not fully general; their format is left open. Directly support content-specific inferences, and defined by the representational role of tracking category membership.

In my view, the two are related by satisfaction; care is needed here. S-concepts apply to individuals: particular dogs, lunches, or runs. G-concepts are higher-order, applying to S-concepts: DOG_G is satisfied by whatever stable categorical representations track the dog category. Importantly, the charge of representational order concerns satisfaction rather than intentionality: thoughts with DOG_G are about dogs. But satisfaction is not constitution: S-concepts satisfy G-concepts, they are not parts out of which G-concepts are built. Thus, which S-concepts satisfy DOG_G may shift, while the G-concept remains the same.

Within a thinker, a G-concept is individuated by its compositional role—its type, in the semanticist’s sense—here, its content-general inferential profile. Across thinkers, two speakers deploy the same G-concept just when the compositional roles of the G-concepts coincide and the S-concepts that satisfy them track the same category. Sharing a G-concept therefore tolerates divergences in content-specific profiles, since these depend on which S-concepts the speakers have accumulated, but not in content-general ones.

Towards the third requirement—aboutness: here, I treat satisfaction as primitive at the architectural level, analogous to Fodor’s Mentalese denotation

or Shea’s label–store connection. The anchoring of a G-concept to a worldly category, via the S-concepts that satisfy it, may ultimately be cashed out along Chomskyan lines (shared cognitive structure plus communicative practice) or Millikian (worldly categories fixed by tracking histories).¹ In any event, G-concepts are usable in reasoning before any such anchoring is established, as §5.3 will show.

4 Stable categorical mechanisms

This section argues that Shea’s neutrality about what identifies a store (§2) is untenable: the individuation condition he himself supplies for concepts presupposes stable categorical mechanisms, in a sense I will make precise. I find that Shea’s picture requires something like S-concepts. The burden of the next section is to argue that S-concepts are insufficient for word meaning.

First, a terminological preliminary. An S-concept, as I use the term, is whatever stably plays a category-tracking role, whether those belonging to the special-purpose systems that Shea discusses, or those postulated by core cognition theorists (Carey 2009; Spelke 2003). ‘Stable categorical mechanism’ names the posit by the work it does, and can be considered independently of its physical realization. To call something an S-concept is to say nothing about how it is organized or addressed in long-term memory, whether by hubs, indices, or distributed networks.

Now, recall Shea’s identity condition (§2): sameness of concept just in case sameness of store. We can grant, with Shea, that multiple special-purpose or domain-specific systems store different information about dogs, for example visual, motor, affective, etc. But what makes all this disparate information *about dogs*, rather than about cats or tables? Shea asserts that “there is a functional reality underpinning the idea that items of information are stored together in memory” (p. 138), but does not commit as to its nature.

He considers the possibility that semantic indexes or hubs play the relevant role, e.g. the hub model of Lambon Ralph et al. [2017], but contends that his approach can succeed with or without committing to such posits (p. 127):

My picture differs from Fodor’s atomism in that working memory labels are not names, i.e. a vehicle type which, across different

¹I endorse a thin, Chomskyan reading on which even paradigm referring expressions like *London* fail to answer to any single set of mind-independent conditions (Chomsky 1986; Chomsky 2000; cf. Harris 2022), with shared content ultimately due to shared psychology. The Millikian tradition closer to Shea’s sympathies takes shared content to arise due to shared psychology *and* a shared world (Millikan 2000).

episodes of thinking, stably provides access to the same body of information; my picture is compatible with there being no index for stored information (p. 152).

If there is no index, the role must be played by distributed connectivity in long-term memory. Shea says as much: storage “may be distributed, with no single index... Even so, the functional fact that items of information are stored together is sufficient to allow us to type-identify labels” (p. 138).

However, such an appeal simply relocates the demand. By Shea’s condition, the WM label I tokened today and that which I tokened last week are instances of the same concept just in case they connected to the same store. Even if some pattern of long-term connectivity can underpin Shea’s “functional reality”, then by his own lights it must do two jobs: make information available to categorization, and make successive labels count as tokens of the same concept. And so, something must make it the case that the store connected to a label today is the very same store as yesterday, robust through changes in information stored. A mechanism that does both jobs is a stable categorical mechanism, however it is physically realized.

Without such mechanisms, nothing in the framework tracks category identity across episodes of thinking, Shea’s own individuation condition is unsatisfiable, and there is nothing stable for words to latch onto. Provided such mechanisms, his official neutrality is forfeit, and Shea’s concepts so far are WM labels connected to what I call S-concepts.

5 Semantic competence requires G-concepts

§4 demanded stable categorical mechanisms. Here, the need is for stable compositional vehicles—symbols, in fact, that persist across episodes, combine freely with others of compatible type, and support content-general inference. As I show, Shea’s combination of temporary WM labels and compositionally limited S-concepts won’t serve. I examine four features of speakers’ grasp of linguistic meaning and highlight in particular the kind of stability each demands: across senses (§5.1), between reasoning and storage (§5.2), across episodes of learning (§5.3), and across modalities of access (§5.4). The G-concept posit meets all four. (1) is vindicated, then, by positing *F*s of precisely this kind.

5.1 Polysemy

Many words exhibit a kind of stable meaning multiplicity which cannot be reduced to indexicality, homophony, vagueness, or generality. For example, the word *lunch* can be used to describe food consumed (stuff) or its consumption (event); *book* can be used to describe a physical volume or its abstract content. These ‘senses’ correspond to distinct categories: lunch can be tasty (stuff) or run long (event); one coffee-stained volume may be the one physical realization of contents shared with thousands of unstained PDFs, etc. Unlike homophones, systematic polysemes alternate between notional classes, and are cross-linguistically robust (Srinivasan and Rabagliati 2015).

Shea adopts Quilty-Dunn’s (2021) treatment of polysemy as context-sensitive retrieval:

In understanding a sentence like ‘the school burnt down’, the concept SCHOOL retrieves information from the store to which it points. Interaction with the other words in the sentence ensures that it is building-related information that is retrieved (school as building, rather than school as organisation or institution, say). (p. 136)

This sounds plausible so far as it goes, but consider co-predication:

- (2) Lunch was delicious and lasted an hour.

In (2), the nominal subject resolves to some stuff, satisfying the first predicate, and to an event, satisfying the second. How would Shea account for this? If comprehension selects just one WM label to bind and connects to LUNCH-STUFF_S, then *lasted an hour* should fail; connecting to LUNCH-EVENT_S, *be delicious* should fail.

Could Shea instead suggest two rounds of selection and binding, garnering distinct WM labels for distinct S-concepts, one per conjunct? He already appeals to a kind of sentence-level “interaction” which can be supplied distinctly by the verb phrases of (2). But such a posit can’t explain polysemy, on pain of failing to differentiate it from homonymy:

- (3) # The bank was flooded and raised interest rates.

Here too we should expect that each predicate can select autonomously and appropriately, having no reason to suppose that the relevant information (river-side, financial institution) is unavailable. Yet, the sentence is unacceptable.²

²Which nouns license which co-predications is lexically specific, and known to speakers

That leaves a single WM label that connects to two stores at once—in effect, a single representation with a complex type. This can work, but only if the label is typed in advance. Shea’s own answer runs the other way: concepts “may be polyvalent. . . depending on the combinatorial structure in which they are deployed” (p. 146). But structure cannot supply what (2) and (3) differ in, since the structures are the same. The difference must lie in a licensing condition carried, in some form, by the nominal itself. A label tokened within the episode cannot invent it; a typed symbol that admits the relevant bindings must persist. Whatever supplies that type structure just is a G-concept, since it composes freely, persists, and is individuated by its type.

With G-concepts in view, *lunch* can be stably associated with the G-concept LUNCH_G , typed so that it associates with both stuff and event S-concepts—a ‘dot-type’, in Pustejovsky’s (1995) sense (see also Asher 2011). The polysemous ‘senses’, then, are grouped under one lexical item in virtue of the shared G-concept. Co-predication structures access the G-concept—a single argument—and apply each of the predicates to it. The intuition that lunch is at once stuff and event reflects a single G-concept being resolved or resolvable to different S-concepts.

5.2 The format of propositional memory

A second demand comes from consideration of storing thoughts. We store propositional information: conclusions we’ve drawn from episodes of reasoning, rumors or facts we’ve heard, beliefs we’ve formed, etc. Shea acknowledges this:

... having learned-through-simulation that the Acme chair will fit in the car, I can store that conclusion as a conceptual belief. Next time I think about the matter, I don’t need to run the simulation again. I can simply recall that *the Acme chair will fit in the car* and reason from there. (p. 131)

But, in what format?

Shea’s own name for what I am calling propositional memory is ‘semantic memory’: “[m]emories stored in the form of explicit conceptual representations” (p. 136), like *Paris is the capital of France*. The question for him is not whether we have such memories, but in the nature of the vehicle that holds

in advance (Ortega-Andrés and Vicente 2019). And this architectural point survives any dispute over whether co-predication involves a type conflict in the first place (Liebesman and Magidor 2017). Even if lunches simply are things that are both food and events, the contrast with (3) is lexical, and minds stably record the difference.

them. Shea’s vehicles are WM labels and representations in special-purpose systems; a stored ‘conceptual representation’ must be built from one or the other. But WM labels can’t store—they are temporary. And the special-purpose representations have the wrong format: Shea individuates these systems precisely by the limited purposes to which their formats may be put (p. 121, quoted above). Formats can span content domains by exploiting structural correspondences between them (e.g., the cognitive map extends from spatial to social content, p. 98), but only so far as those correspondences hold.

Special-purpose formats employ compositional principles of their own, but the content of stored propositional memory is integrative:

The picture I’m building integrates information from lots of different special-purpose systems: sensorimotor knowledge of ways of preparing ingredients, olfactory and gustatory simulation of potential dishes, affective-evaluative responses to the imagined results; a spatial map [...]; motoric knowledge of the effort cost of various potential actions; and semantic knowledge of the monetary cost of potential ingredients. The suppositional scenario is multi-system and holistic. *Concepts allow the thinker to manipulate and recombine specific elements of these scenarios.* (p. 124, emphasis added)

We commit conclusions drawn from such scenarios to memory. Suppose you came to stably believe that the three red vehicles blocking the intersection were rented from the same company. Storing that belief involves holding together, in one representation, information about color, number, space, and institutions. No special-purpose format handles all four, and no one format’s principles are expected to reach the other three. Whatever stores the belief must be general-purpose by Shea’s lights; but unlike his WM labels, it must also persist.

Shea himself forecloses the possibility that the belief can be reassembled at recall: “I don’t need to run the simulation again. I can simply recall” it (p. 131). On the Hybrid view, propositional memories can be stored as complexes of G-concepts, in their general-purpose format, leaving the vocabularies of reasoning and storage the same. Verifying, perceiving, and acting may very well require accessing S-concepts, but S-concepts cannot substitute for the general-purpose vocabulary.

5.3 Fast versus extended mapping

A third demand emerges from language development. Consider hearing *Bo has a glee* with a novel noun, with no other information given about what

gleebies are. Despite such limitations, you can nevertheless immediately reason:

- (4) If gleebs are expensive, then Bo has something expensive.
- (5) All of Bo's things are green; therefore, Bo has a green thing.

A fresh word figures in inference without prior content; yet, *gleeb* is taken to mean the same thing where it occurs, and to compose with the rest on the presumption that it picks out a category. Shea's framework allows for a WM label to be assigned to a (near-)empty store, and to compose freely based on a syntactic type assigned in the moment. What happens when the same word is heard again?

Here the dissociation between 'fast mapping'—taking up a word's form and semantic class from a single exposure—and 'extended mapping'—the slow filling-in of what the word applies to—is important. In Carey and Bartlett [1978]'s original demonstration of fast mapping, preschoolers heard the novel item *chromium* (meaning OLIVE GREEN) in a contrastive frame—*bring me the chromium tray; not the red one, the chromium one*. A week later, any child that recalled anything about *chromium* recalled that it was a color word, usually without knowing which specific color it picked out. Carey [2010] finds the same patterning for verbal number: for months before they appreciate its specific cardinal value, children understand *two* to be a quantifier that contrasts with *one*. This pattern of 'class membership first, precise denotation later' across domains strains Shea's account.

Since WM labels are temporary, Shea has few resources to explain what children retain about *chromium* or understand about *two* in the early days. He might point to Horst and Samuelson [2008]'s much-discussed finding that 24-month-olds show only fragile retention of newly mapped words. However, the fragility inheres just in the link to the specific category, or S-concept—children recall the word's form and compositional potency. Indeed, the ability to pick out the intended referent in the moment and to retain the specific link dissociates across development, with the number of new words that show successful selection exceeding how many can be successfully retained [Bion et al., 2013]. A basic or 'skeletal' knowledge consisting of the word and an associated conceptual class can be fixed quickly, while acquiring further and more specific information is slow-going.

This profile is unsurprising on the Hybrid view. Fast mapping generates an appropriately typed G-concept, immediately compositionally available in virtue of its type. This is all that is needed for a G-concept to figure in content-general inferences like (4) and (5). An initial association links the new item to some S-conceptual class. Across episodes, the generated symbol is retained;

piecemeal knowledge of restrictions on its satisfiers can be further associated from subsequent learning context. This separate, later, and more effortful process of extended mapping reflects the association of S-conceptual material with a stable vehicle that was fast-mapped to support composition.

5.4 Modality of access

Speakers who acquire their lexicon very differently highlight a fourth demand. If speakers’ linguistic understanding is generally anchored just to information in special-purpose systems, we might expect significant divergences between congenitally blind and sighted speakers’ lexicons. But we don’t.

Landau and Gleitman [1985] documented a congenitally blind child producing visual verbs like *look* and *see* early, with appropriate use by 36 months. Later work has shown that the productive vocabulary of congenitally blind toddlers is “statistically indistinguishable” from sighted children’s, both in semantic category and perceptual modality [Campbell et al., 2024]. Even more, blind people construct extensive, intuitive causal theories of color, light, and visual perception that go well beyond the meanings of individual words [Wang et al., 2026]. On Shea’s picture, this patterning is hard to understand. Whatever the initial state of their visual system, the blind learner lacks a visual input channel that would trigger or support their formation of visual S-concepts. And yet their lexical entries for visual words arrive on the same schedule as for sighted learners.

The Hybrid view explains this convergence. The G-concept RED_G , for example, has a general-purpose, amodal format, and is individuated by its compositional role. The inferences that fall out—content-general ones in virtue of its type-theoretic structure, content-specific in virtue of semantic class alone (§5.3), such as that red is a color—should be shared across the two populations. What varies is the route to more deeply content-specific knowledge. Sighted speakers can access such information via visually accessed S-concepts, while blind speakers may lean more heavily on linguistically transmitted propositional memory, stored in G-conceptual format (§5.2). The invariant compositional vehicle, across modalities, is the G-concept.

6 Conclusion

I have argued that speakers’ grasp of linguistic meaning makes demands on cognitive architecture that, so far, Shea’s framework doesn’t meet. Semantic competence requires stable, freely compositional vehicles to serve as consis-

tent targets for word meanings—what the Hybrid view calls G-concepts. The empirical landscape concentrates the demand for stability at four points: polysemy resolution, propositional storage, fast-to-extended mapping, and different modalities of informational access. G-concepts are distinct from but intimately related to the S-concepts that underwrite categorization. A pluralist could in principle propose some device to meet each distinct demand, at the obvious cost of multiplying posits where the G-concept would do.

Translation between natural languages illustrates the architectural point. An English speaker producing *dog* and a Finnish speaker producing *koira* are, by hypothesis, in cognitive states mediating the same content. Shea’s WM labels don’t underwrite this identity, as he grants (p. 139), since they are temporary even within speakers. Drawing stable word-to-S-concept links won’t do it either, since English $dog \rightarrow DOG_S$ and Finnish $koira \rightarrow DOG_S$ are distinct relations, even if they terminate at the same S-concept. Yet the two speakers must share something that composes into the thoughts both are capable of expressing;³ the relevant type-identity resides at the G-level.

These moves can be resisted, but not without cost. Rejecting (1) incurs the burden of accounting for the empirical data I have surveyed via psycholinguistic accounts that don’t yet exist. Replacing my characterization of the compositional vehicle with some new form of distributed connectivity in long-term memory would challenge the format, but not the needed functional role. I haven’t debated such possibilities; rather, I have worked to reveal the cost of keeping Shea’s framework entire.

My claims are robust across metasemantic options. Shea himself favorably cites Pietroski [2018] for the idea that word meanings are instructions for accessing concepts (pp. 146–147), a third option alongside the mere ‘involvement’ and identity readings of (1). The four stability demands I have highlighted bear on whatever-it-is that speakers necessarily access when deploying knowledge of meaning in comprehension, inference, and translation. Whether word meanings are G-concepts, whether grasping meaning merely involves them, or whether meanings are instructions for accessing them—the architectural profile of the conceptual targets matches that of the G-concept.

The Hybrid view’s distinctive payoff is a two-level architecture with concepts indeed at the interface. It consists of stable, type-individuated G-concepts composing at an order above the S-conceptual satisfiers that anchor them to the world. Working memory remains the arena of thought: it holds

³If such a link is stable across occasions, capable of compositional relations, and individuated by what it applies to, then it is not a mere link but a representation in its own right, with the hallmarks of a G-concept.

and composes tokens of G-concepts, from which the S-concepts satisfying them can be activated. Lacking G-concepts, Shea’s framework delivers only compositionality without stability, or stability without compositionality. The facts demand both.

My argument has been architectural; with Shea, I leave open questions about how such an architecture could arise—in particular, whether G-concepts precede or emerge with properly linguistic cognition.

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References

- Nicholas Asher. *Lexical Meaning in Context: A Web of Words*. Cambridge University Press, Cambridge, UK, 2011.
- Ricardo A. H. Bion, Arielle Borovsky, and Anne Fernald. Fast mapping, slow learning: Disambiguation of novel word–object mappings in relation to vocabulary learning at 18, 24, and 30 months. *Cognition*, 126(1):39–53, 2013.
- Elisabeth Camp. The generality constraint and categorial restrictions. *The Philosophical Quarterly*, 54(215):209–231, 2004. doi: 10.1111/j.0031-8094.2004.00348.x.
- Erin Campbell, Robyn Casillas, and Erika Bergelson. The role of vision in the acquisition of words: Vocabulary development in blind toddlers. *Developmental Science*, 27(4):e13475, 2024.
- Susan Carey. *The Origin of Concepts*. Oxford University Press, Oxford, 2009.
- Susan Carey. Beyond fast mapping. *Language Learning and Development*, 6(3):184–205, 2010. doi: 10.1080/15475441.2010.484379.
- Susan Carey and Elsa Bartlett. Acquiring a single new word. In *Proceedings of the Stanford Child Language Conference*, volume 15, pages 17–29, 1978.

- Noam Chomsky. *Knowledge of language*. Praeger Publishers, New York, New York, 1986.
- Noam Chomsky. *New Horizons in the Study of Language and Mind*. Cambridge University Press, Cambridge, UK, 2000.
- Gareth Evans. *Varieties of Reference*. Oxford University Press, 1982.
- Jerry A. Fodor. *Concepts: Where Cognitive Science Went Wrong*. Oxford University Press, 1998.
- Brendan Gillon. Ambiguity, Indeterminacy, Deixis, and Vagueness. In Stephen Davis and Brendan Gillon, editors, *Semantics: A reader*, chapter 10, pages 157–187. Oxford University Press, Oxford, UK, 2004.
- Daniel W. Harris. Semantics without semantic content. *Mind & Language*, 37(3):304–326, 2022. doi: 10.1111/mila.12290.
- Jessica S. Horst and Larissa K. Samuelson. Fast mapping but poor retention by 24-month-old infants. *Infancy*, 13(2):128–157, 2008.
- Matthew A. Lambon Ralph, Elizabeth Jefferies, Karalyn Patterson, and Timothy T. Rogers. The neural and computational bases of semantic cognition. *Nature Reviews Neuroscience*, 18:42–55, 2017.
- Barbara Landau and Lila R. Gleitman. *Language and Experience: Evidence from the Blind Child*. Harvard University Press, Cambridge, MA, 1985.
- David Liebesman and Ofra Magidor. Copredication and property inheritance. *Philosophical Issues*, 27:131–155, 2017.
- Ruth Garrett Millikan. *On Clear and Confused Ideas*. Cambridge University Press, 2000.
- Marina Ortega-Andrés and Agustín Vicente. Polysemy and co-predication. *Glossa: A Journal of General Linguistics*, 4(1):1–23, 2019. doi: 10.5334/gjgl.564. Art. 1.
- Paul Pietroski. *Conjoining meanings: Semantics without truth values*. Oxford University Press, Oxford, UK, 2018.
- James Pustejovsky. *The Generative Lexicon*. MIT Press, Cambridge, MA, 1995.

- Jake Quilty-Dunn. Polysemy and thought: Toward a generative theory of concepts. *Mind & Language*, 36(1):158–185, 2021.
- Nicholas Shea. *Concepts at the Interface*. Oxford University Press, 2024.
- Elizabeth Spelke. What makes us smart? Core knowledge and natural language. In Dedre Gentner and Susan Goldin-Meadow, editors, *Language in Mind: Advances in the Study of Language and Thought*, pages 277–311. MIT Press, Cambridge, MA, 2003.
- Mahesh Srinivasan and Hugh Rabagliati. How concepts and conventions structure the lexicon: Cross-linguistic evidence from polysemy. *Lingua*, 157:124–152, 2015. doi: 10.1016/j.lingua.2014.12.004.
- Ziwen Wang, Akshi, Sophia Keil, Judy Kim, and Marina Bedny. Constructing meaning from language: Visual knowledge in people born blind and in large language models. *Annual Review of Linguistics*, 12:295–323, 2026. doi: 10.1146/annurev-linguistics-011724-121432.